

ESIT-2026 Challenge - Task 2

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Family A — Relabel: denoise the labels

Give the model a *trusted* target for each forget sample, then fine-tune. The error against the *original* label then equals the corruption.

1. For each forget sample i , find its $k = 10$ nearest **retain** neighbours in the 64-dim PCA of $|\text{CSI}|$.
2. Replace its label with their mean position: $\hat{\mathbf{y}}_i^{\text{corr}} = \frac{1}{k} \sum_{j \in \text{kNN}_{\mathcal{R}}(i)} \mathbf{p}_j$ (retain labels untouched).
3. Fine-tune on the corrected targets.

Family B — Activation-trajectory divergence

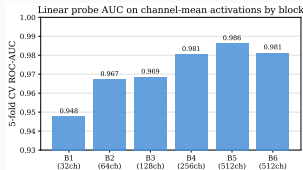
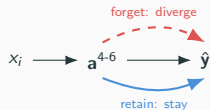
Motivation: the CNN *already* separates forget from retain internally (probe AUC **0.986** at block 5), yet parameter-space dampening (SSD) *fails* (Fisher importance entangled).

So act in activation space.

Method (vs. frozen teacher f_{θ_0}):

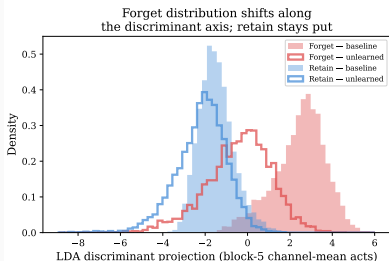
- **Diverge** forget pre-ReLU acts off-trajectory.
- **Anchor** retain post-ReLU acts to the teacher.
- + kNN forget-position loss (transferable signal).

Real representational change, not just a relabel.

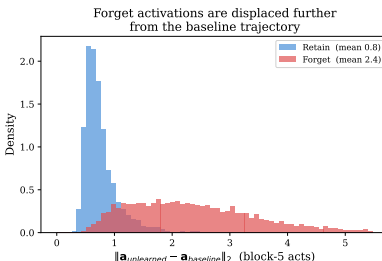


Seeing the divergence in activation space

Effect of activation-trajectory divergence on block-5 representations



Left — where they sit. One axis (LDA) is fit on the *teacher* to best tell forget from retain at block 5; every sample is projected onto it. Filled = before, outline = after. The **forget** cloud slides ~ 3 units off its baseline location; **retain** does not move.



Right — how far each moved. Per-sample $\|a_{\text{after}} - a_{\text{before}}\|_2$. **Retain** barely shifts (mean 0.8); **forget** travels $3\times$ further (mean 2.4), the two hardly overlap.

Relabelling only remaps the output; here the **internal representation itself moves and only for forget**.

The small ensemble

The two families make *complementary* errors: relabel gives a clean transferable signal; divergence detaches the representation. Averaging their per-sample errors before the detector yields a **more separable** error distribution.

